

# SAS PROJECT

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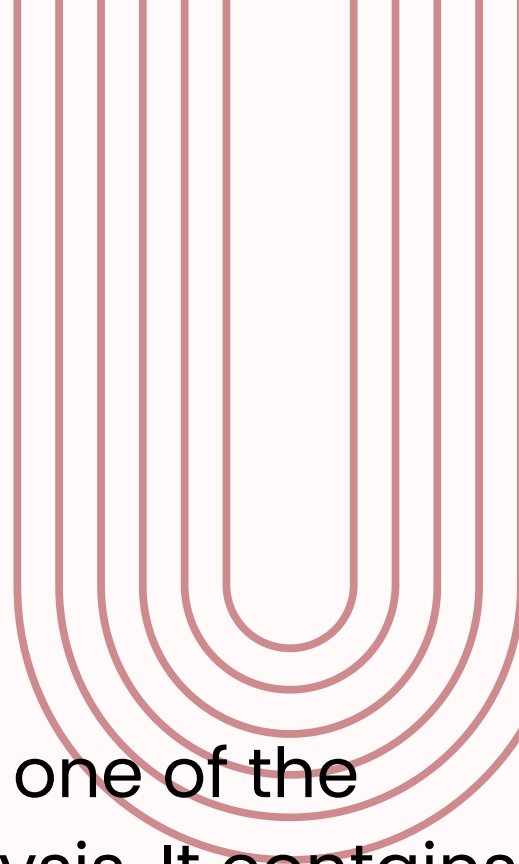
ID: 2401248124 شهد احمد فرغلي احمد G2



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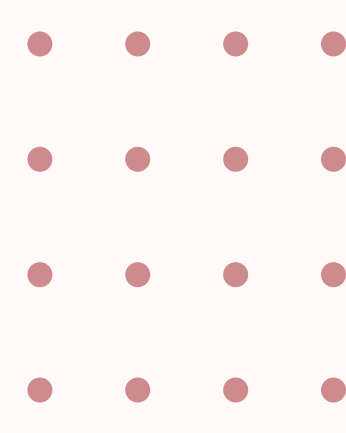
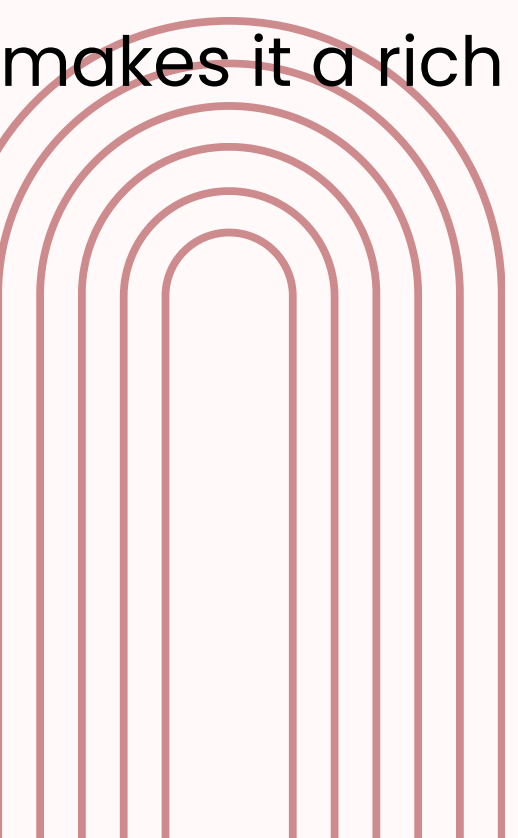
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## **Why I Chose This Dataset:**

This dataset was chosen because healthcare is one of the most data-rich and impactful domains for analysis. It contains nearly 10,000 real-world-style patient records, offering a large enough sample to draw meaningful statistical conclusions. The dataset is particularly interesting because it combines both patient demographics and clinical outcomes, allowing us to explore relationships between personal attributes (like age, gender, and blood type) and medical results. Furthermore, the presence of a clear target variable — Test Results — makes it well-suited for both descriptive analysis and potential predictive modeling. The diversity of variables, spanning financial data, medical conditions, and hospital logistics, makes it a rich candidate for multi-angle exploration.



# What Was Done in the Code:

## 1. Data Import & Overview

The CSV file was imported into SAS using PROC IMPORT, and PROC CONTENTS was used to inspect the structure of the dataset — column names, data types, and variable count.

## 2. Descriptive Statistics

PROC MEANS was applied to the three numerical variables: Age, Billing Amount, and Room Number — reporting count, mean, standard deviation, min, max, and median for each.

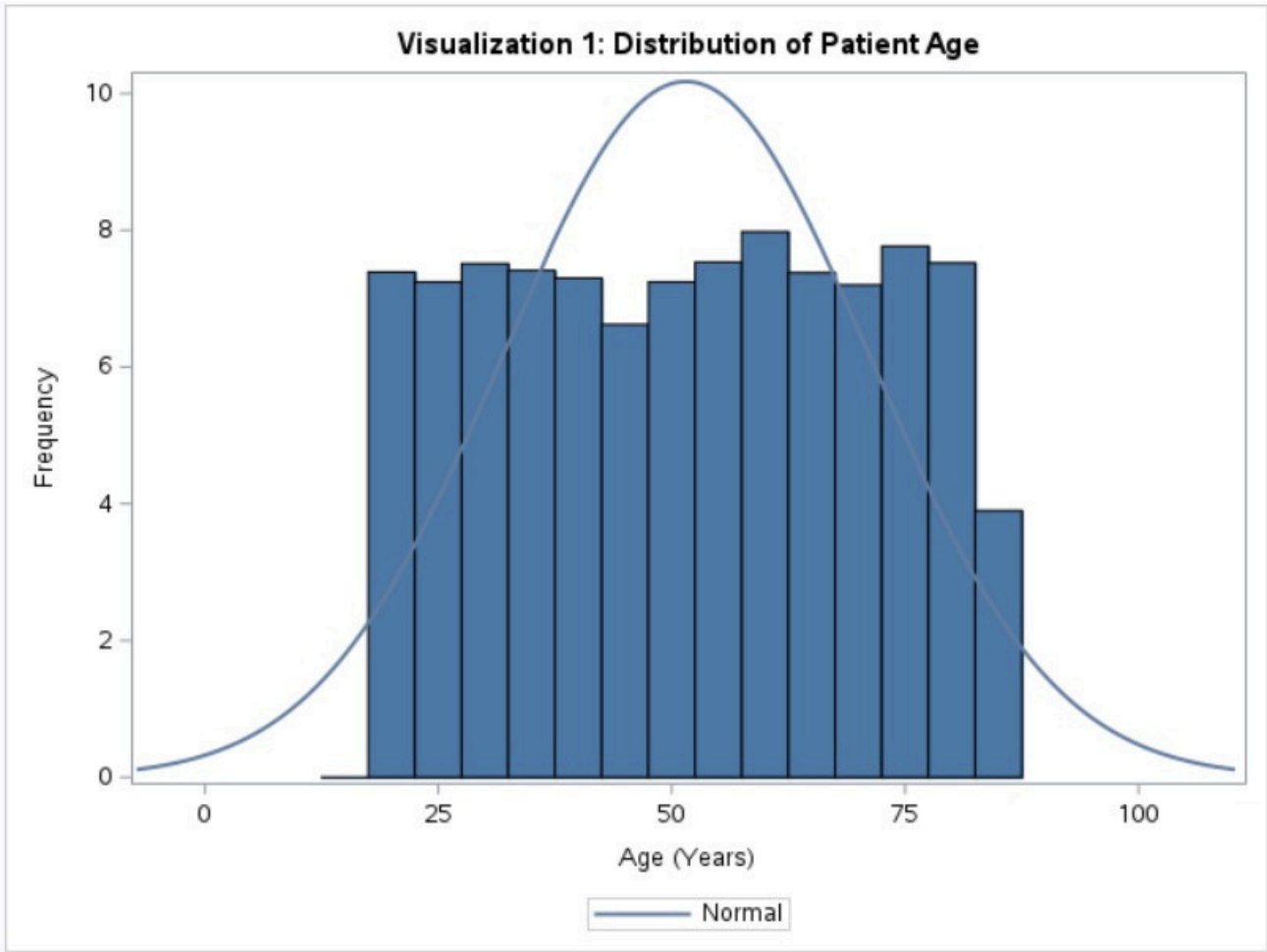
## 3. Frequency Tables

PROC FREQ was used on all categorical variables: Gender, Blood Type, Medical Condition, Admission Type, Medication, Test Results, and Insurance Provider — to see the distribution and count of each category.

# FIRST PLOT: DISTRIBUTION OF PATIENT AGE

This histogram (with an overlaid normal distribution curve) shows the age distribution of patients in the dataset.

- Key Findings:
- Age Range: Patients range from approximately 18 to 85 years old.
  - Central Tendency: The distribution is roughly normal (bell-shaped), with the mean age around 51.5 years and a median of 52 years.
  - Peak Frequency: The highest frequencies occur in the 25–75 age range, with particularly high counts in the 40–70 bracket. This suggests the dataset primarily captures adult and senior patients.
  - Tails: Fewer patients at the extremes (very young adults under 25 or those nearing 85+).
  - Implication: The patient population is broadly representative of a typical hospital's adult demographic, with a slight right skew but overall good spread across working-age and elderly groups

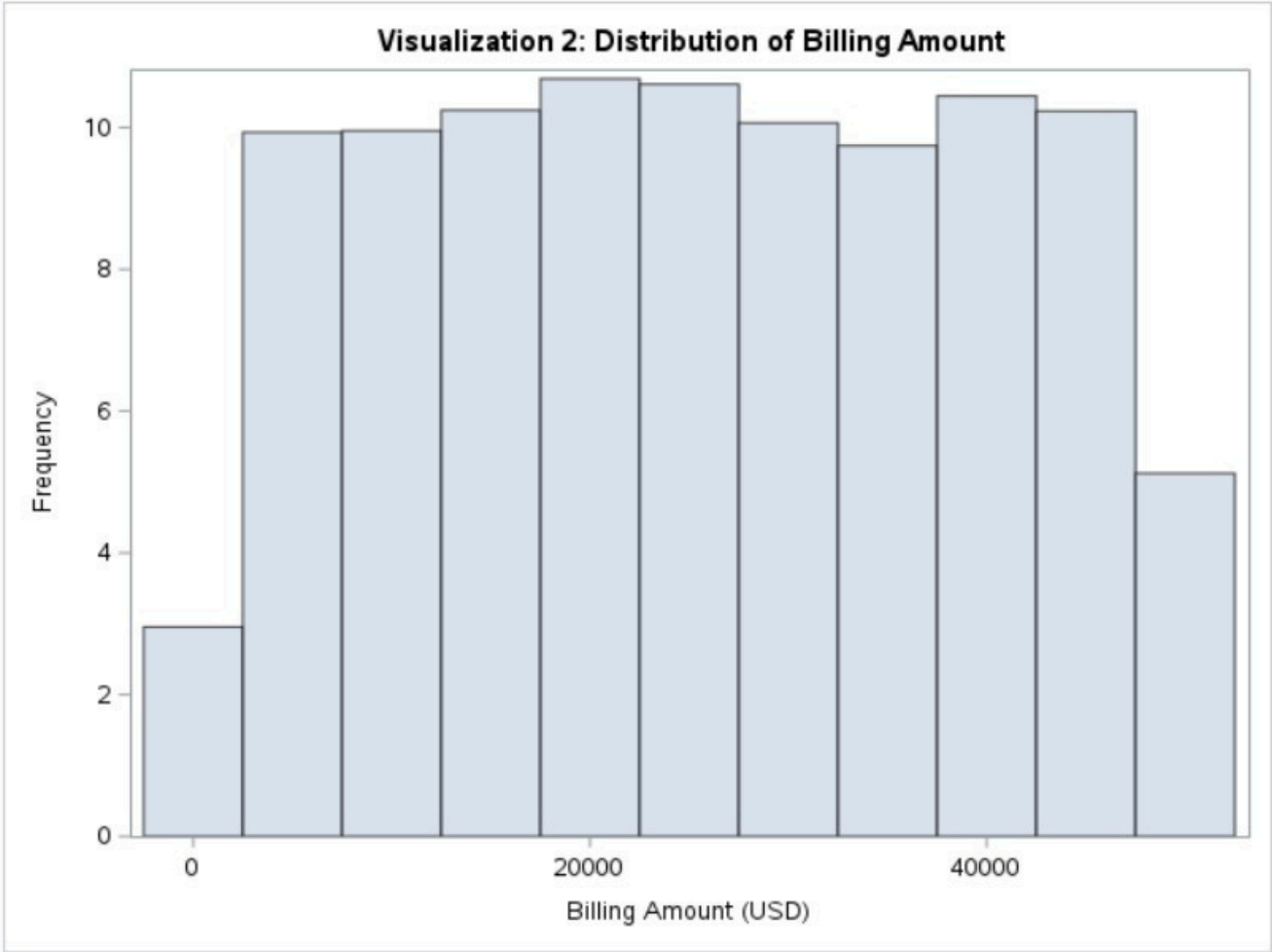


# SECOND PLOT :DISTRIBUTION OF BILLING AMOUNT

This histogram displays the distribution of billing amounts (in USD).

Key Findings:

- Range: Billing amounts span from roughly \$1,000 to \$50,000.
- Shape: The distribution is relatively uniform across most bins (especially \$10,000–\$45,000), with slightly lower frequency at the very low end (< \$5,000) and a drop at the highest bin (above ~\$45,000).
- Average: The mean billing amount is approximately \$25,576, with a median of \$25,310.
- Variability: Standard deviation is high (~\$14,073), indicating significant variation in costs depending on conditions, length of stay, treatments, etc.
- Implication: Most patients incur mid-to-high bills, with no extreme outliers visible in this view. Costs appear fairly evenly distributed rather than heavily skewed toward low or high amounts

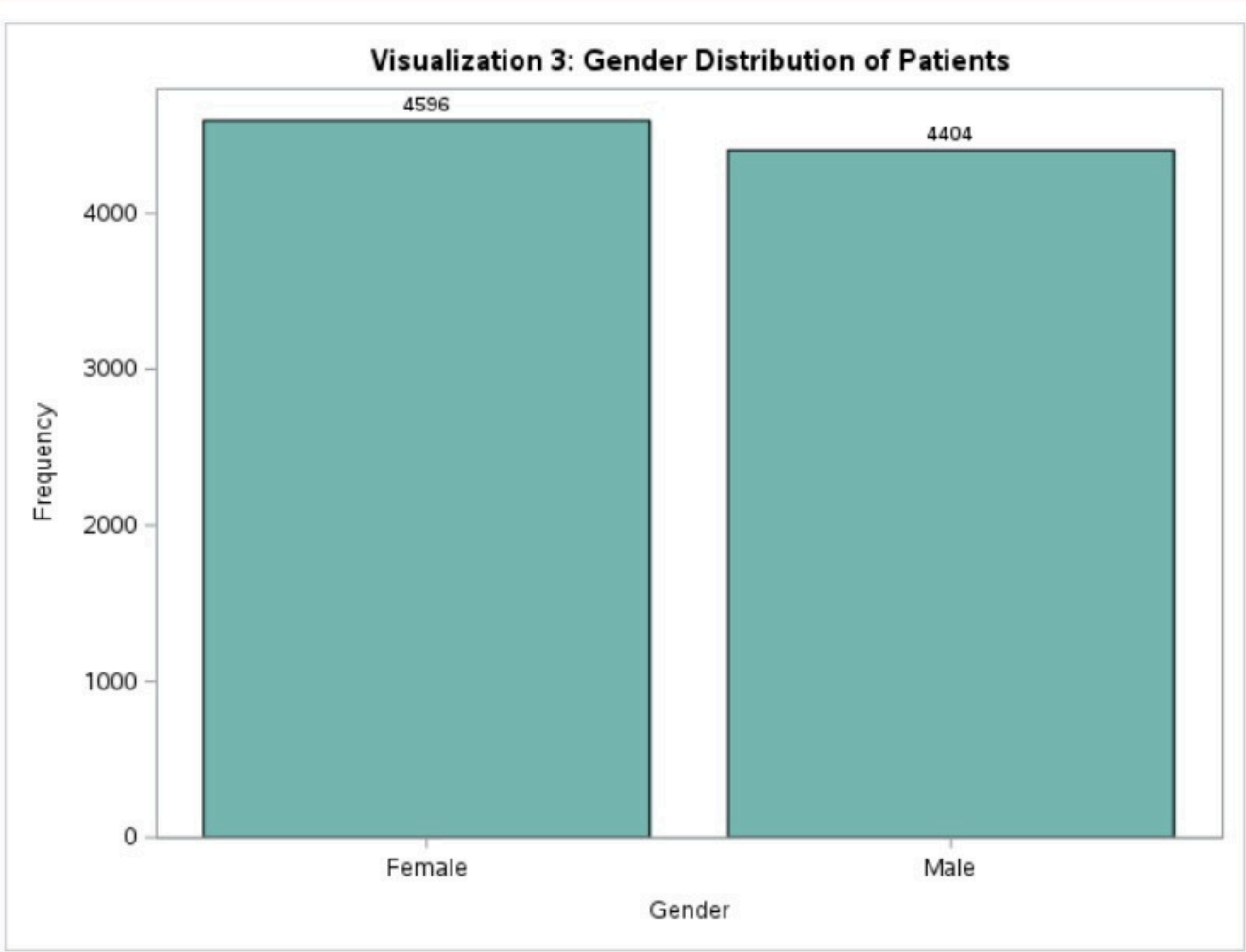


# THIRD PLOT:GENDER DISTRIBUTION OF PATIENTS

This bar chart compares the number of female and male patients.

Key Findings:

- Female: 4,596 patients
  - Male: 4,404 patients
  - Unknown/Missing: ~1,000 records (inferred from total dataset size)
- Implication: There is a slight female majority (roughly 51% female vs. 49% male among known genders). The genders are nearly balanced, which is useful for analyzing potential disparities in treatment, costs, or outcomes by gender.



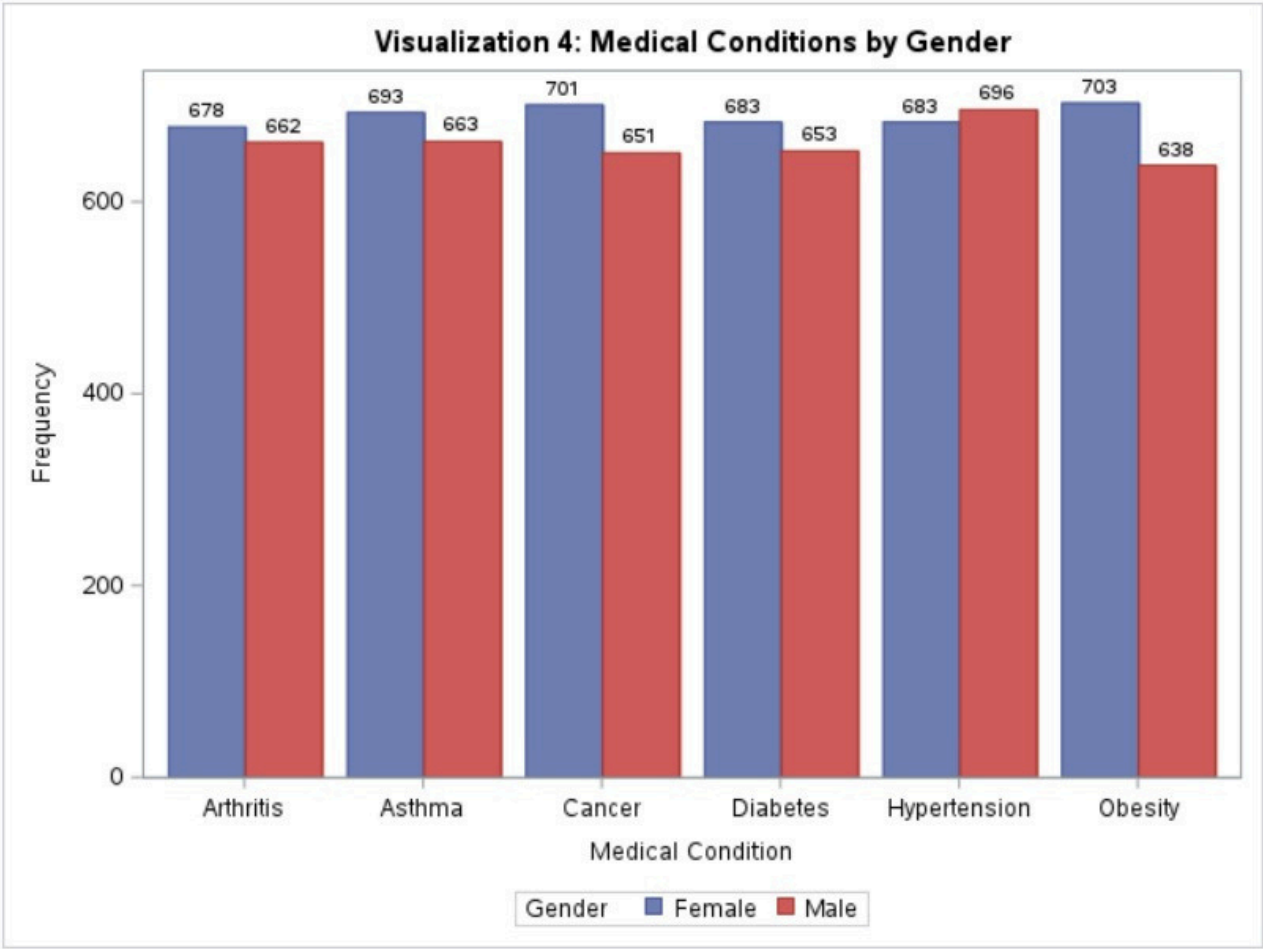


# FOURTH PLOT:MEDICAL CONDITIONS BY GENDER

This grouped bar chart breaks down the six main medical conditions by gender (Female in blue, Male in red).

Key Findings:

- Most Common Conditions (overall counts):
  - o Hypertension: ~1,531
  - o Cancer: ~1,525
  - o Asthma: ~1,511
  - o Arthritis: ~1,486
  - o Obesity: ~1,482
  - o Diabetes: ~1,465





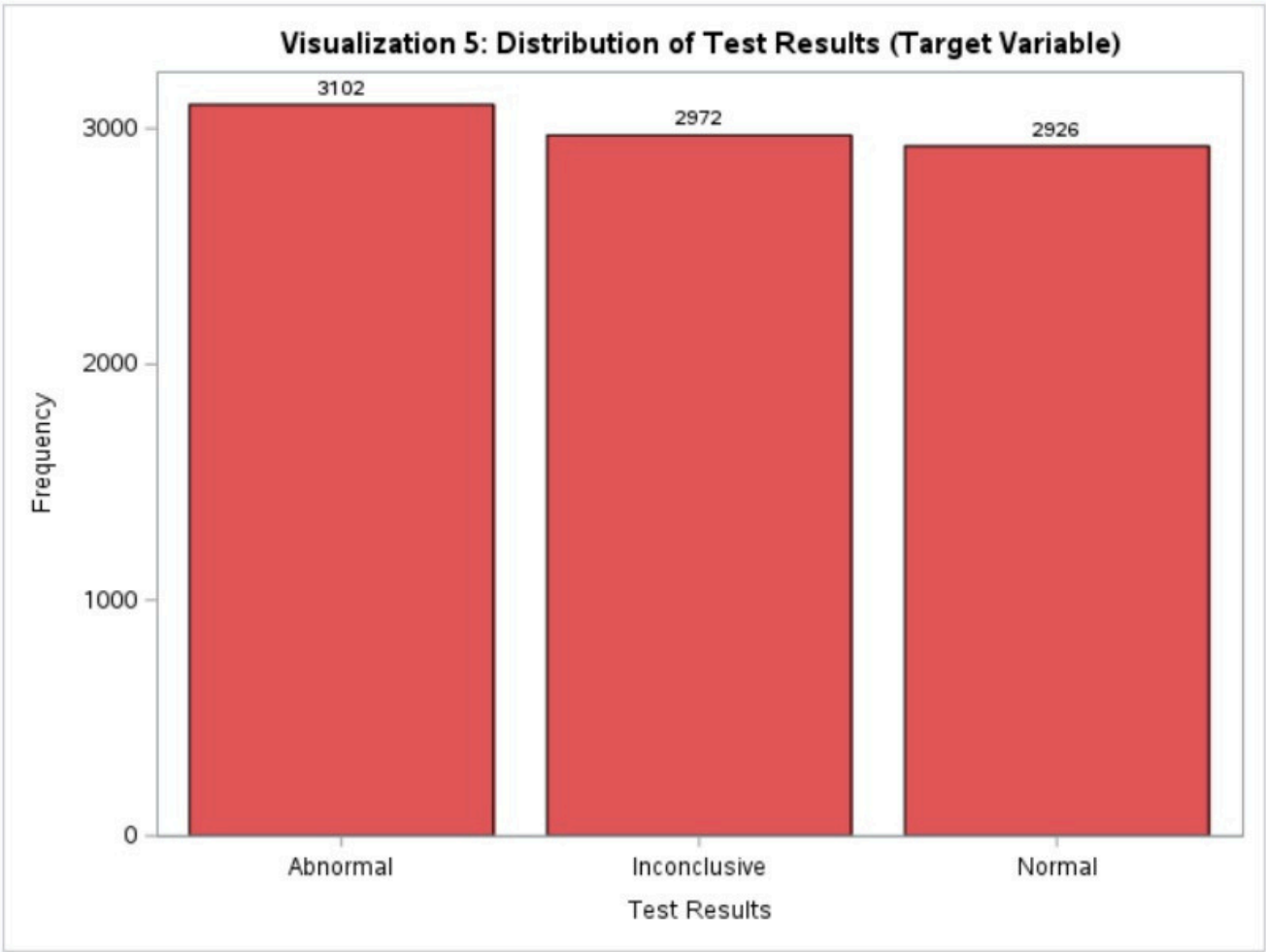
# FIFTH PLOT: DISTRIBUTION OF TEST RESULTS

This bar chart shows the frequency of each test result category.

Key Findings:

- Abnormal: 3,102 (highest)
- Inconclusive: 2,972
- Normal: 2,926

Implication: The target variable (Test Results) is quite balanced, with only minor differences between categories. Abnormal results are slightly more common, but all three outcomes are nearly equally represented. This is ideal for classification modeling as there is no severe class imbalance.



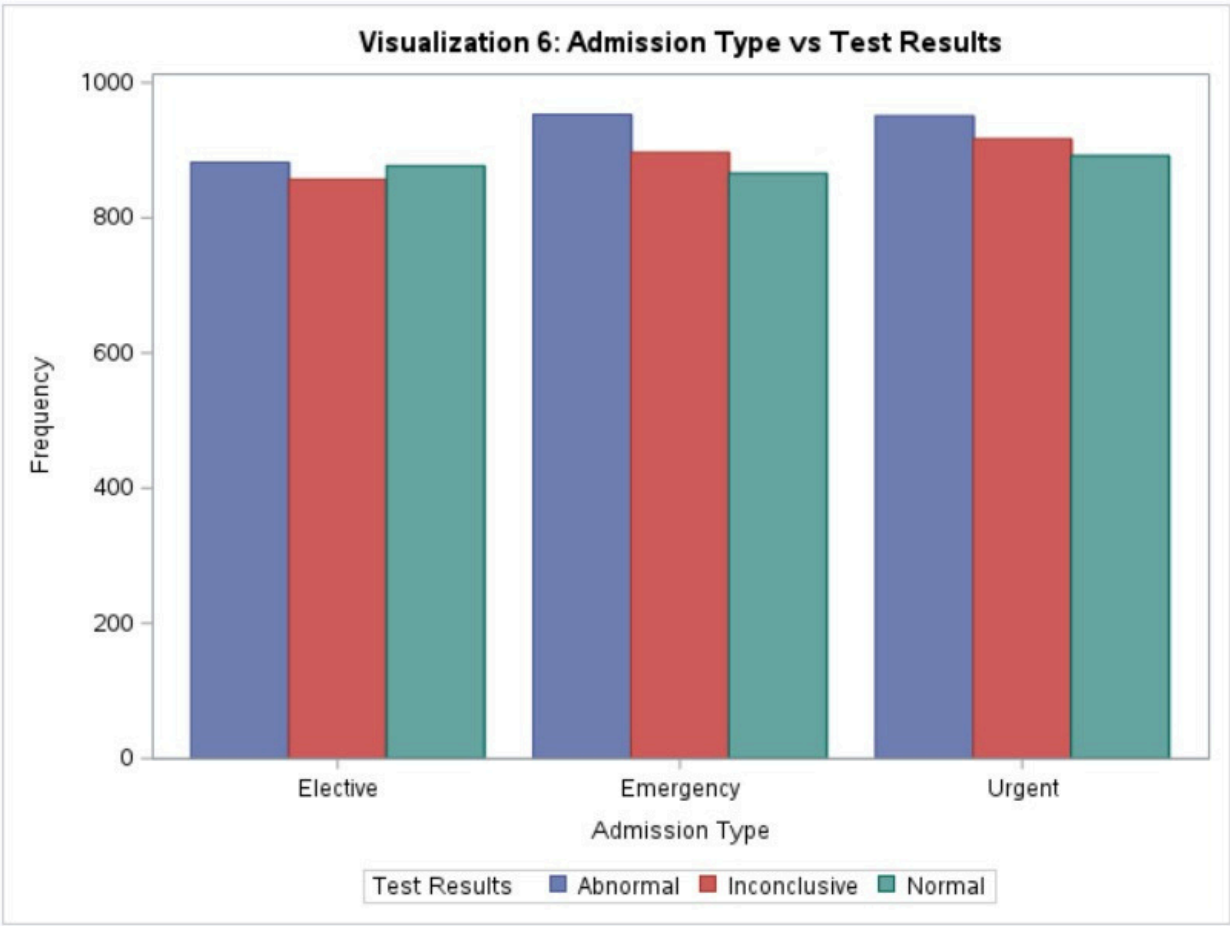
# SIXTH PLOT :ADMISSION TYPE VS RESULT TYPE

This grouped bar chart compares test results across Elective, Emergency, and Urgent admissions.

Key Findings:

- Elective: Roughly balanced (Abnormal ~882, Inconclusive ~857, Normal ~877).
- Emergency: Slightly higher Abnormal (953), followed by Inconclusive (897) and Normal (866).
- Urgent: Also slightly higher Abnormal (951), with Inconclusive (917) and Normal (892).

Implication: Across all admission types, Abnormal test results are marginally more frequent. The distribution of test outcomes is quite similar regardless of admission type, suggesting that admission urgency does not strongly dictate test result categories.



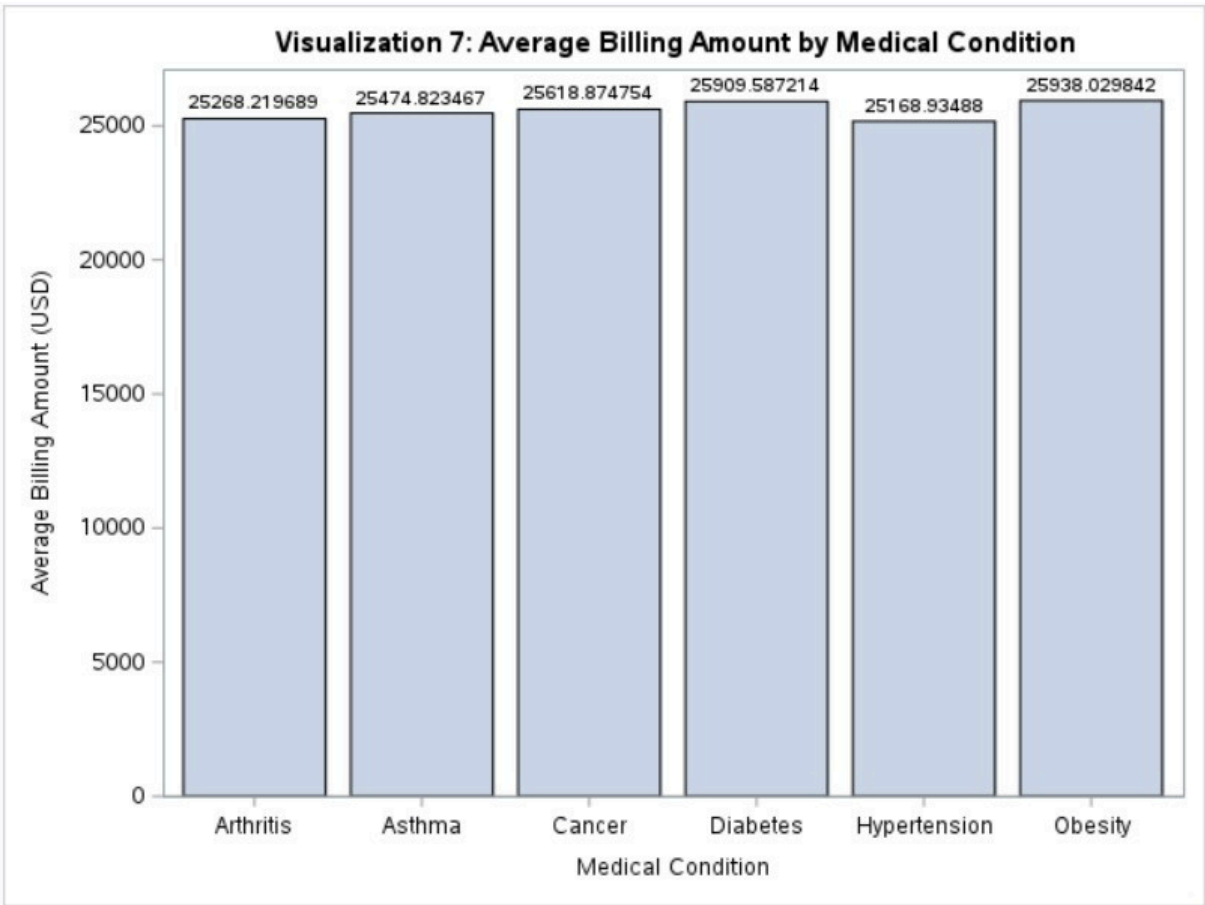
# SEVENTH PLOT :AVERAGE BILLING AMOUNT BY MEDICAL CONDITION

This bar chart displays the mean billing amount for each medical condition.

Key Findings:

- Diabetes: \$25,910 (highest)
- Obesity: \$25,938
- Cancer: \$25,619
- Asthma: \$25,475
- Arthritis: \$25,268• Hypertension: \$25,169 (lowest)

Implication: Average billing amounts are very consistent across conditions (all hovering tightly around \$25,000–\$26,000). Differences are small, suggesting that billing is driven more by other factors (e.g., length of stay, procedures, or complications) than by the specific diagnosis alone.

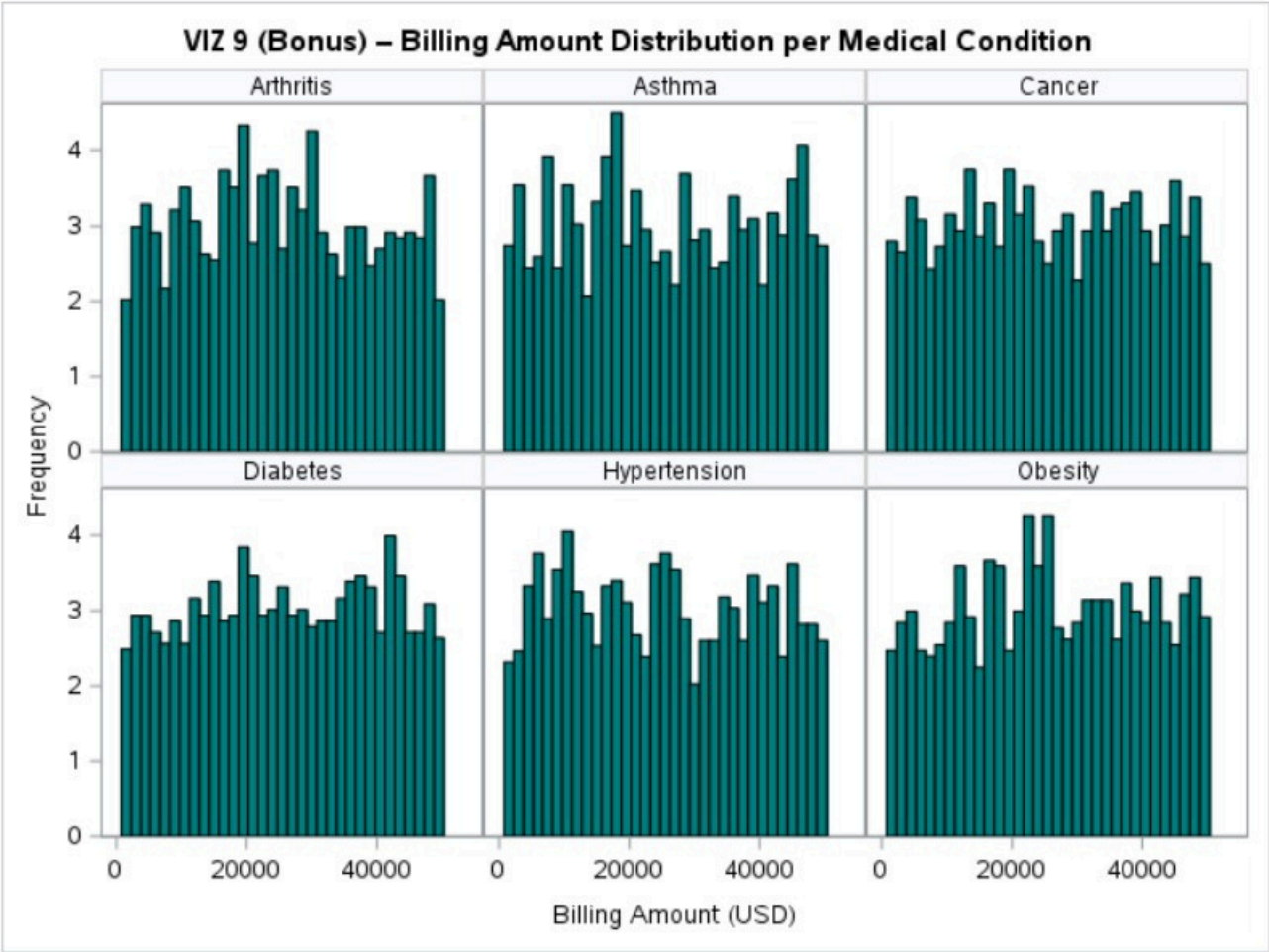


# NINTH PLOT: BILLING AMOUNT DISTRIBUTION PER MEDICAL CONDITION

## Key Findings:

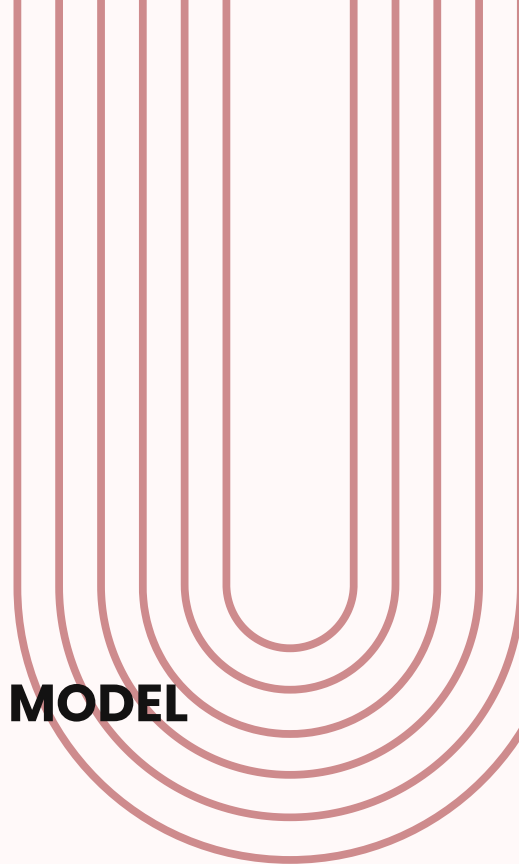
- All conditions (Arthritis, Asthma, Cancer, Diabetes, Hypertension, Obesity) display very similar distributions.
- Broad spread from low to high bills (~\$0 to \$50,000).
- Roughly uniform or slightly multimodal patterns, with peaks in the mid-range.
- No condition stands out with dramatically different cost patterns.

Implication: Billing amount distributions are highly consistent regardless of the medical condition. This reinforces that costs are not strongly tied to specific diagnoses in this dataset.





# FEATURE ENGINEERING



## 4. FEATURE ENGINEERING

**NEW FEATURES WERE CREATED TO IMPROVE MODEL PERFORMANCE:**

### 4.1 LENGTH OF STAY

**CALCULATED AS:**

**DISCHARGE DATE – ADMISSION DATE**

**REPRESENTS HOSPITAL STAY DURATION.**

### 4.2 COST PER DAY

**BILLING AMOUNT / LENGTH OF STAY**

**MEASURES FINANCIAL INTENSITY OF TREATMENT.**

### 4.3 AGE GROUP

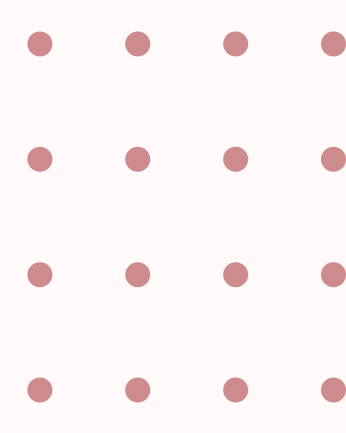
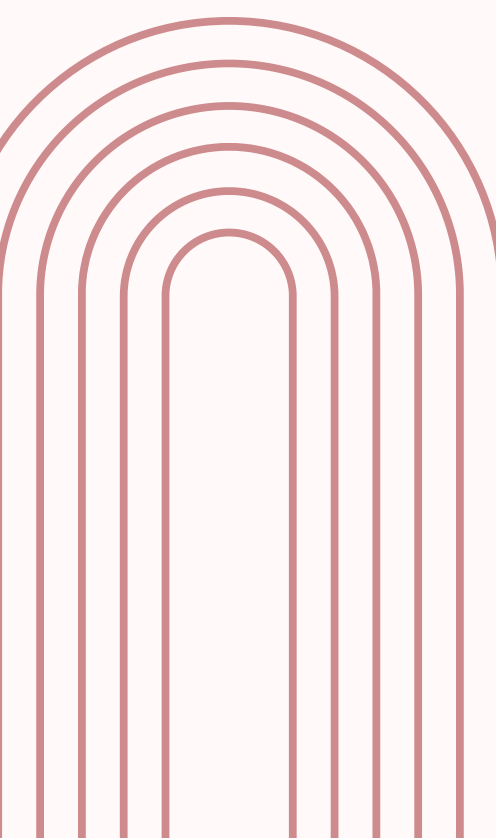
**PATIENTS WERE CATEGORIZED INTO:**

**CHILD (<18)**

**YOUNG (18–39)**

**MIDDLE (40–59)**

**SENIOR (60+)**



# FEATURE ENGINEERING

## 4.4 RISK INDICATORS

**BINARY FLAGS WERE CREATED:**

**EMERGENCY ADMISSION**

**CHRONIC CONDITIONS (DIABETES, CANCER, HEART DISEASE)**

## 4.5 TARGET VARIABLE (HIGH RISK)

**A RULE-BASED TARGET WAS DEFINED:**

**A PATIENT IS CONSIDERED HIGH RISK IF:**

**AGE > 60**

**OR CHRONIC DISEASE EXISTS**

**OR COST PER DAY > 1000**

**FINAL RULE:**

**IF SCORE  $\geq 2 \rightarrow$  HIGH RISK (1)**

**OTHERWISE  $\rightarrow$  LOW RISK (0)**

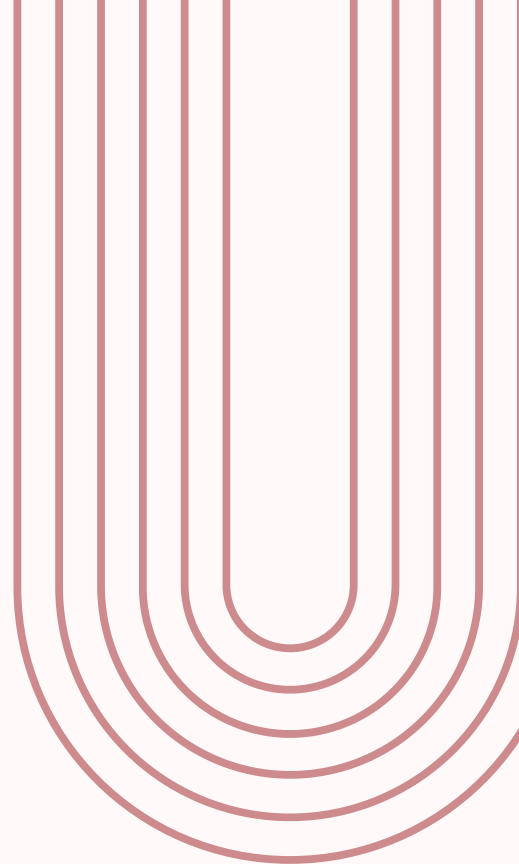
## 4.6 BILLING LEVEL

**BILLING AMOUNT WAS CATEGORIZED INTO:**

**LOW**

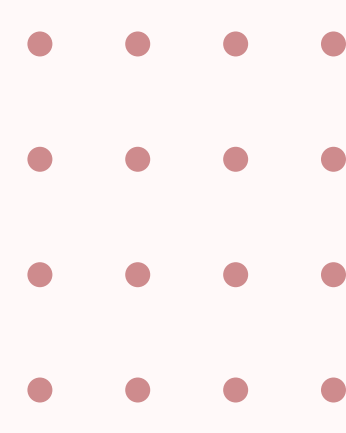
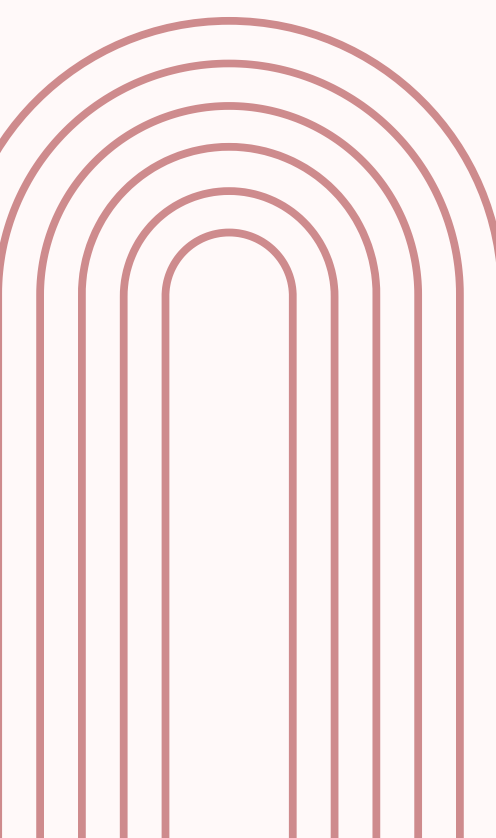
**MEDIUM**

**HIGH (BASED ON QUARTILES Q1 AND Q3)**

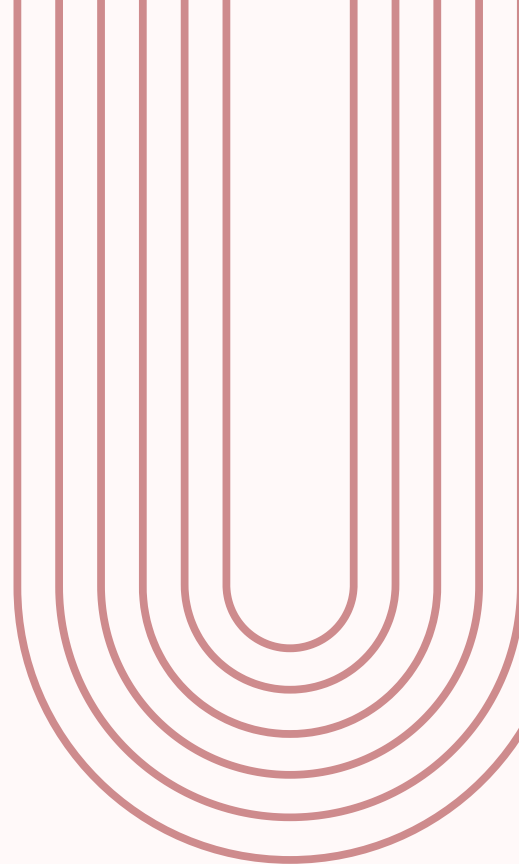


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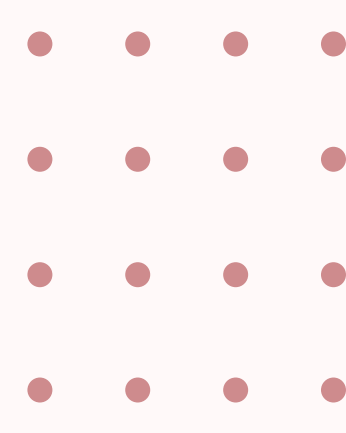
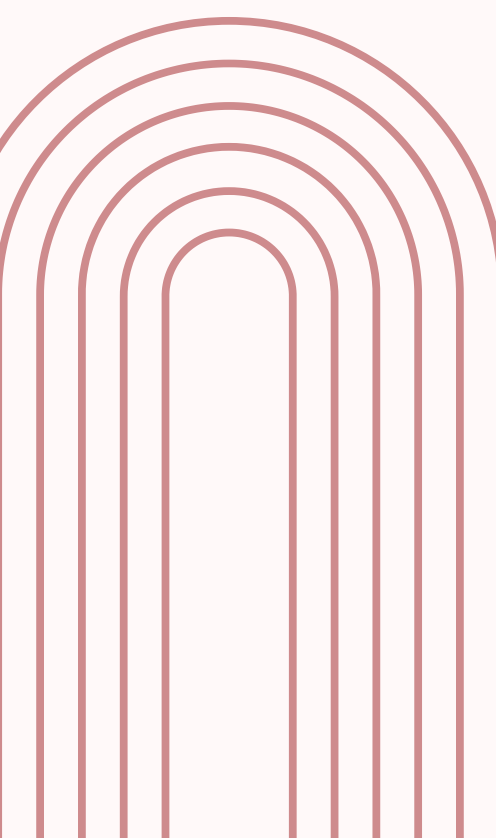




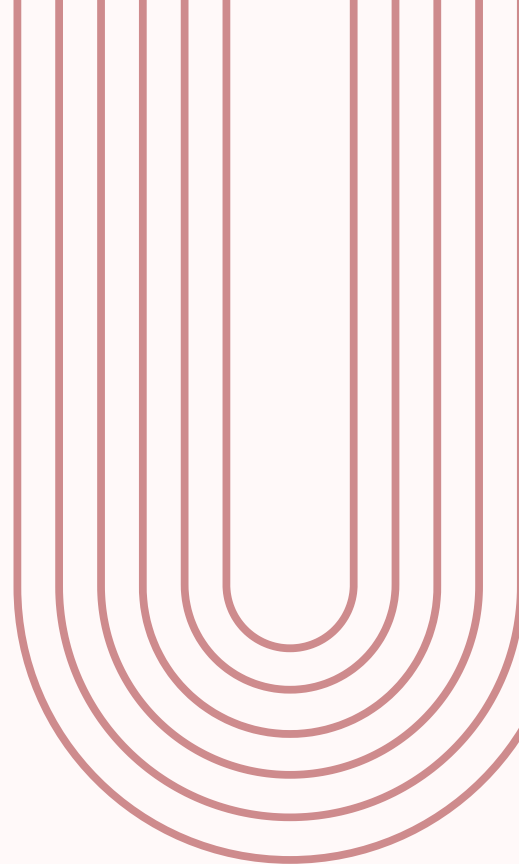


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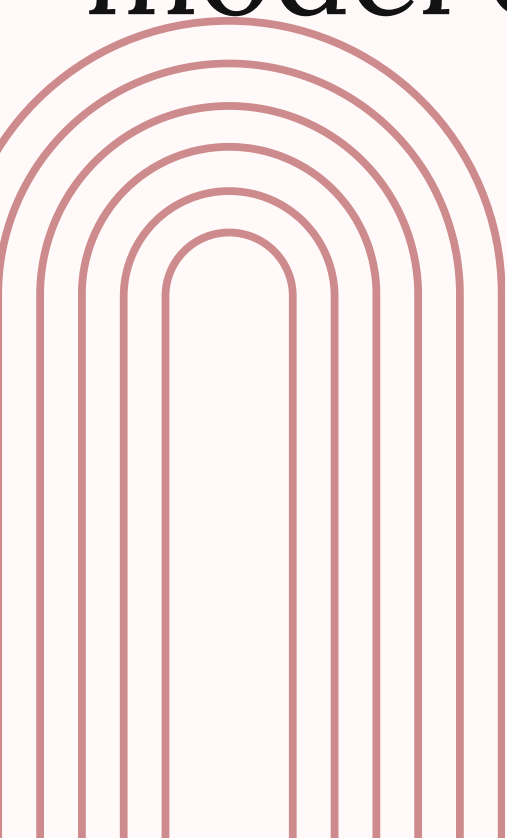
# DATA PREPROCESSING

## 3.1 Train/Test Split

The dataset was split into:

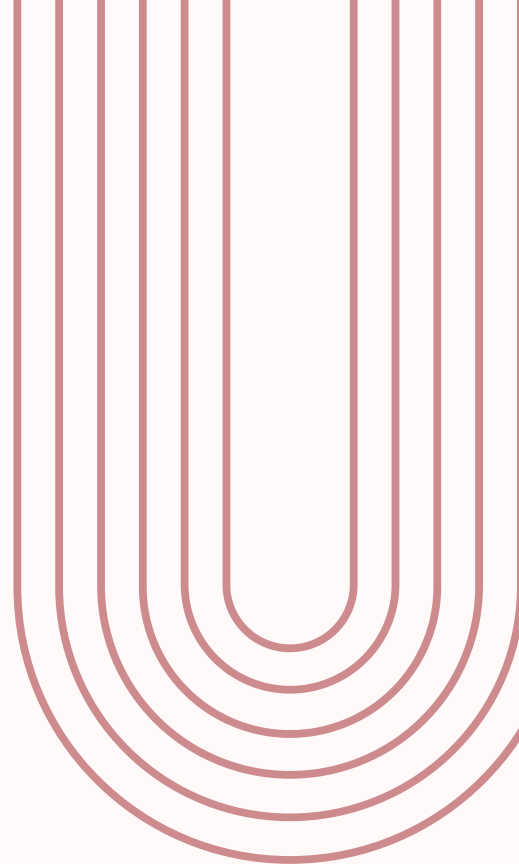
- 70% Training data
- 30% Testing data

This ensures fair evaluation of the model on unseen data.





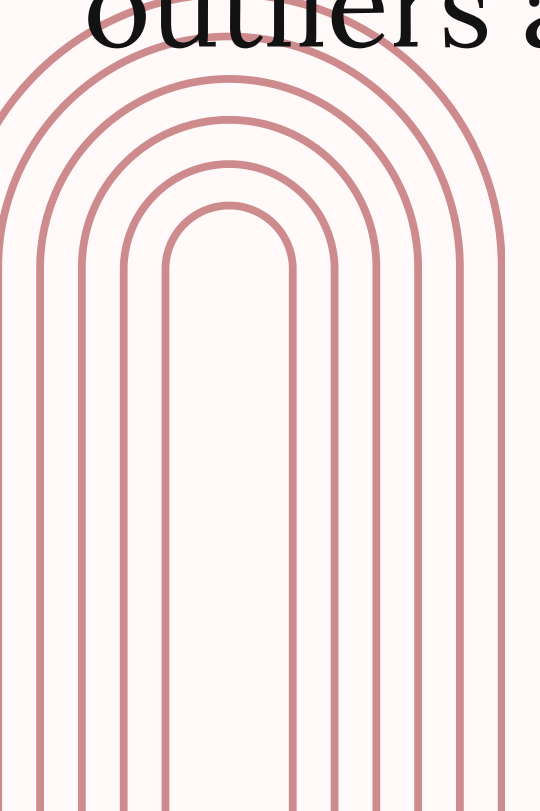
# MISSING VALUES HANDLING

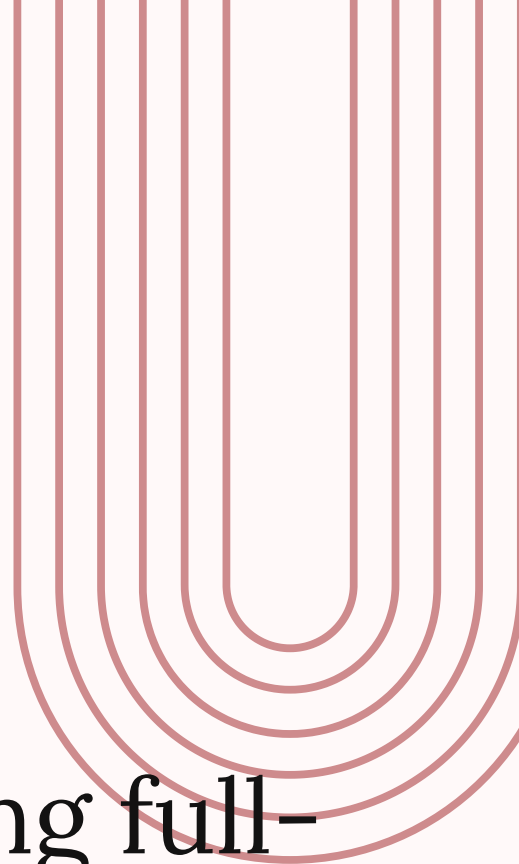


Missing values were treated as follows:

- Categorical variables → replaced with Mode (most frequent value)
- Numerical variables → replaced with Median

This approach reduces the effect of outliers and keeps data realistic.



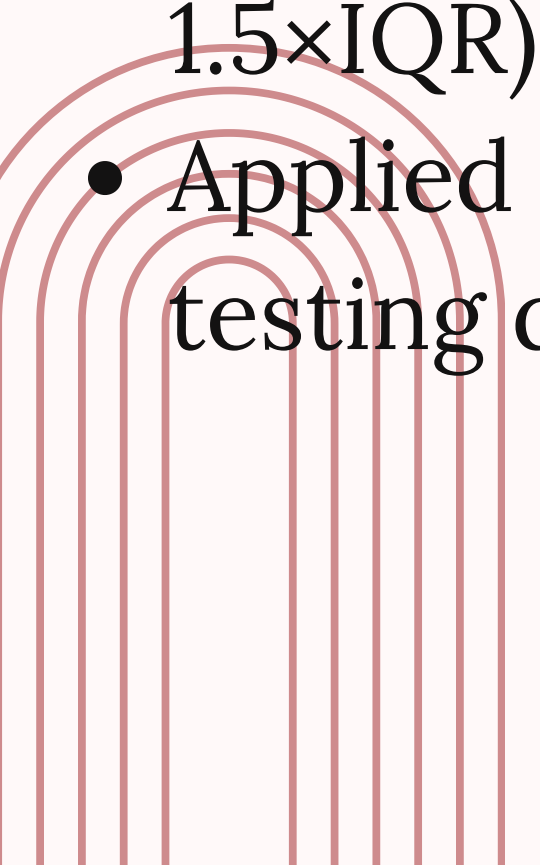


# DUPLICATE REMOVAL

Duplicate rows were removed using full-column comparison to ensure data quality and avoid bias in modeling.

## Outlier Handling

Outliers were detected using the IQR method (Interquartile Range):

- Values outside  $(Q1 - 1.5 \times IQR)$  or  $(Q3 + 1.5 \times IQR)$  were removed.
  - Applied consistently on both training and testing datasets.
- 
- 

# MODEL

## 1. Logistic Regression

- Predicts probability of High Risk
- Uses statistical relationship between variables
- Suitable for binary classification problems

## 2. Decision Tree (HPSPLIT)

- Splits data into decision rules
- Easy to interpret
- Uses validation split (30%) for evaluation

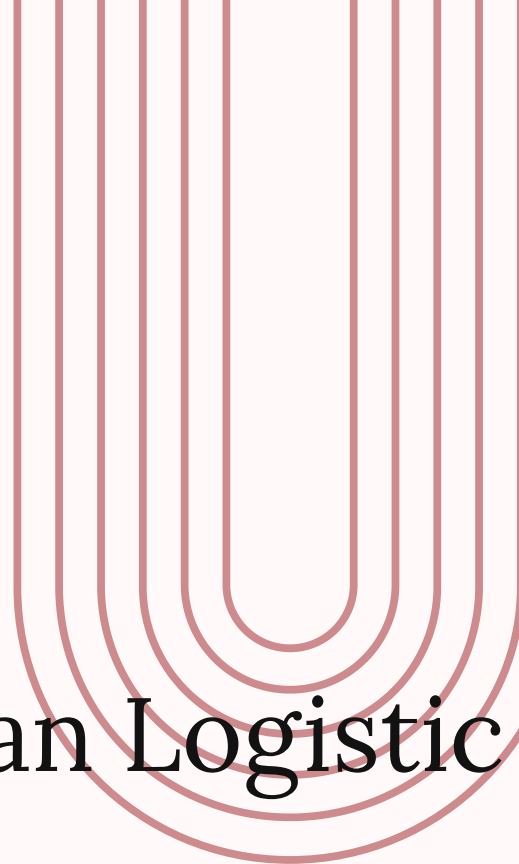
## 5. Model Evaluation

### Logistic Regression

- Accuracy  $\approx 92.7\%$
- Strong performance on test data
- Stable and reliable predictions



## Decision Tree

- Accuracy  $\approx 79.9\%$
  - Slightly lower performance than Logistic Regression
  - Better interpretability but less accuracy
- 

## 6. Model Comparison

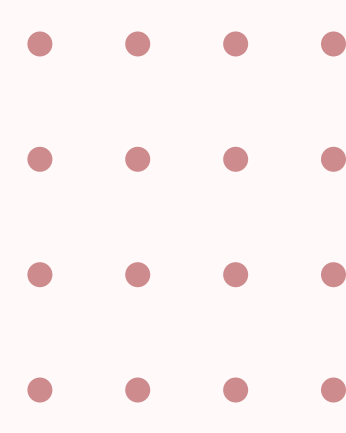
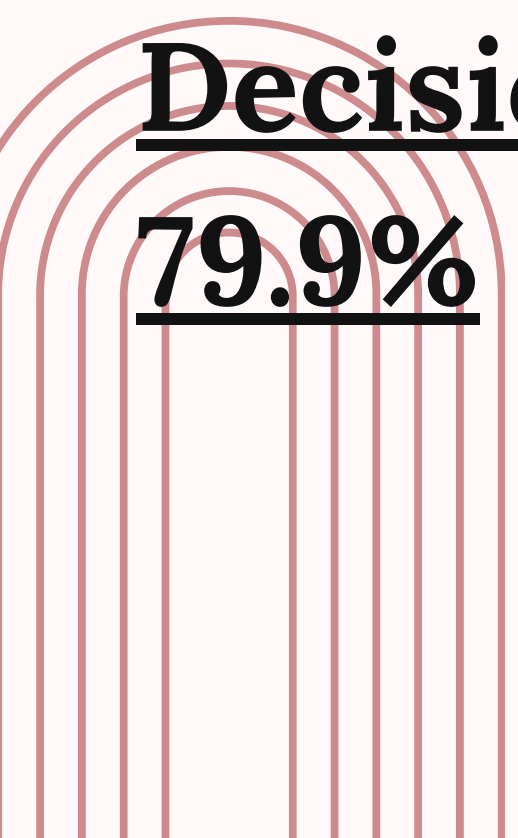
Model Accuracy

Logistic Regression

92.7%

Decision Tree

79.9%



## **7. Conclusion**

- **Logistic Regression achieved the highest accuracy and performed better overall.**
- **Decision Tree provided interpretable rules but lower predictive performance.**